THE SHADOW OBSTRUCTION TOWER OF A CHIRAL ALGEBRA AND THE TRANSFERRED L_∞ TOWER

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ABSTRACT. Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve and let $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ be its modular convolution dg Lie algebra, with bar-intrinsic Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$. In the verified chirally Koszul genus-zero cyclic setting, after a cyclic contraction of $\mathfrak{g}^{\mathrm{mod},(0)}_{\mathcal{A}}$ onto its shadow cohomology is fixed, the finite-order shadow projections of $\Theta_{\mathcal{A}}$ coincide with the transferred L_∞ bracket evaluations on the projected Maurer–Cartan element. Degree two is the curvature input; degrees $r \geq 3$ are the formality obstruction evaluations. For the standard families where the transfer data are constructed, this comparison identifies the shadow termination degree with the L_∞ obstruction level.

1. Introduction

1.1. Formality and its obstructions. A dg Lie algebra $\mathfrak{g}$ is *L_∞-formal* if its minimal L_∞-model is L_∞-isomorphic to the strict graded Lie algebra $(H^*(\mathfrak{g}), 0, [-,-]_H)$. For a fixed contraction, the transferred brackets ℓ_r^{tr} are representatives; the obstruction is their class in the appropriate deformation complex after all lower obstructions have been killed. Kontsevich's formality theorem [3] proved that polyvector fields on a smooth manifold form an L_∞-formal dg Lie algebra, giving deformation quantization of Poisson structures; Tamarkin [8] recovered the result operadically from formality of the little disks. When formality fails, obstructions are recorded in cohomology classes $o_3, o_4, \ldots$: at degree r the class o_r measures whether ℓ_r^{tr} vanishes on cohomology; if $o_r = 0$, a homotopy kills ℓ_r^{tr} and the next obstruction lives one degree higher. The sequence $(o_r)_{r\geq 3}$ is the *formality obstruction tower*; see Loday–Vallette [5, Ch. 10].

1.2. The shadow tower of a chiral algebra. On a smooth projective curve X, a chiral algebra $\mathcal{A}$ [1] has a bar complex $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, where $\bar{\mathcal{A}} = \ker(\epsilon)$ is the augmentation ideal, with deconcatenation coproduct and differential built from the Arnold form $\eta_{ij} = d\log(z_i - z_j)$ on the Fulton–MacPherson compactification of the configuration space. The modular convolution dg Lie algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is the pronilpotent dg Lie algebra whose differential encodes graph sums over stable curves in $\overline{\mathcal{M}}_{g,n}$. The bar-intrinsic element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0 \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$ satisfies $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ and packages the full genus expansion of $\mathcal{A}$ [6, MC2 and modular convolution theorem]. Its finite-degree truncations $\Theta^{\leq r}_{\mathcal{A}}$ define the *shadow obstruction tower*

$$\mathrm{Sh}_r(\mathcal{A}) \;:=\; -\nabla_H^{-1}\big([\Theta^{\leq r-1}_{\mathcal{A}}, \Theta^{\leq r-1}_{\mathcal{A}}]^{(0)}_r\big), \qquad r \geq 2,$$

where the bracket is genus-zero graph composition and ∇_H is the Hessian covariant derivative on the shadow algebra $\mathcal{A}^{\mathrm{sh}}$. The first three shadows are $\mathrm{Sh}_2(\mathcal{A}) = \kappa(\mathcal{A})$ (the modular characteristic [6, Theorem D]), $\mathrm{Sh}_3(\mathcal{A}) = \mathfrak{C}(\mathcal{A})$ (cubic shadow), and $\mathrm{Sh}_4(\mathcal{A}) = \mathfrak{Q}(\mathcal{A})$ (quartic shadow).

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1.3. **Shadow and transfer.** The two towers are built in different complexes. The formality tower is the transferred cyclic Chevalley tower of the genus-zero convolution dg Lie algebra. The shadow tower is the scalar projection of the bar-intrinsic Maurer–Cartan recursion. The comparison is a theorem only after the genus-zero cyclic contraction is fixed.

Theorem 1.1 (Genus-zero shadow–formality comparison). *Let $\mathcal{A}$ be a chirally Koszul standard algebra for which the genus-zero cyclic convolution complex, shadow projection, and cyclic contraction (ι, π, h) are constructed. Assume the cyclic stable-tree/HPL comparison lemma: after cyclic symmetrisation, the genus-zero stable-tree graph sum and the Kadeishvili–Merkulov transfer sum have identical signs, automorphism weights, propagators, and vertex decorations. For every $r \geq 2$,*

$$\mathrm{Sh}_r(\mathcal{A}) \;=\; \ell_r^{(0),\mathrm{tr}}\big(\pi\Theta_{\mathcal{A}}^{\leq r-1}, \dots, \pi\Theta_{\mathcal{A}}^{\leq r-1}\big) \quad \text{in } \mathcal{A}^{\mathrm{sh}}_{r,0},$$

relative to the fixed cyclic contraction. Degree 2 is the curvature term. For $r \geq 3$ this is the transferred formality obstruction evaluated on the truncated Maurer–Cartan class.

The proof has two ingredients. The cases $r = 2, 3, 4$ are the explicit computations of Proposition 4.1. For $r \geq 5$, the Kadeishvili–Merkulov tree formula [2, 7] on the strict genus-zero convolution algebra projects, after cyclic symmetrisation, to the same stable-tree graph sum that defines the shadow coefficient.

1.4. **Consequences.** Theorem 1.1 identifies the shadow termination degree $r_{\max}(\mathcal{A})$ with the L_∞ obstruction level $f_\infty(\mathcal{A})$ of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$ for the verified standard cyclic families. Operation-level equality with a transferred A_∞-depth requires a separate nondegeneracy statement. The four shadow-depth classes are realized by Heisenberg (class **G**, $r_{\max} = 2$, formal), affine Kac–Moody (class **L**, $r_{\max} = 3$), $\beta\gamma$ (class **C**, $r_{\max} = 4$), and Virasoro (class **M**, $r_{\max} = \infty$, intrinsically non-formal); see Section 7.

Scope. A verified chirally Koszul standard algebra is one for which the genus-zero cyclic convolution complex, the shadow cohomology projection, and a cyclic contraction (ι, π, h) have been constructed. All comparison statements below are relative to this fixed transfer datum. They do not assert formality of the full two-coloured $\mathrm{SC}^{\mathrm{ch,top}}$ operad, nor of the Calabi–Yau stage-one factorisation algebra before Stage-2 specialisation.

Conventions. Grading is cohomological with $|d| = +1$; desuspension lowers degree, $|s^{-1}v| = |v| - 1$. Bar complexes use the augmentation ideal: $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$. The propagator $d\log E(z, w)$ on the bar complex has weight one. All statements are at the level of cohomology classes unless otherwise noted.

2. The shadow obstruction tower

2.1. **The bar complex and its convolution algebra.** Let X be a smooth projective curve over $\mathbb{C}$ and let $\mathcal{A}$ be a chiral algebra on X [1]. The ordered chiral bar complex $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}}) = \bigoplus_{n\geq 0}(s^{-1}\bar{\mathcal{A}})^{\otimes n}$ carries the deconcatenation coproduct, with differential assembled from $\eta_{ij} = d\log(z_i - z_j)$ and the chiral product $\mu\colon\; j_*j^*(\mathcal{A} \boxtimes \mathcal{A}) \to \Delta_*\mathcal{A}$. The *modular convolution dg Lie algebra* is

$$\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} \;:=\; \mathrm{Hom}^{\mathrm{cyc}}\big(\bar{B}(\mathcal{A}),\, \mathcal{A}\big) \otimes \mathsf{G},$$

where $\mathsf{G} = \bigoplus_{g\geq 0} \mathsf{G}^{(g)}$ is the coefficient algebra generated by stable graphs on $\overline{\mathcal{M}}_{g,n}$. The genus filtration gives $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} = \bigoplus_{g\geq 0}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})_g$; the genus-zero part $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)} = \mathrm{Hom}^{\mathrm{cyc}}(\bar{B}(\mathcal{A}), \mathcal{A}) \otimes \mathsf{G}^{(0)}$ is a strict dg Lie algebra with bracket $[-,-]^{(0)}$ from tree composition, since genus-zero stable graphs are trees.

2.2. **The Maurer–Cartan element and degree truncations.** The differential on $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ splits as $D = d_0 + D_{\mathrm{int}}$, with d_0 linear and D_{int} the interacting piece encoding the chiral product and the Arnold form. The difference $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0 \in \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is a Maurer–Cartan element [6, MC2 bar-intrinsic construction]:

$$D\Theta_{\mathcal{A}} + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0.$$

The element $\Theta_{\mathcal{A}}$ is bar-intrinsic. Write

$$F_{\le r}\bar{B}(\mathcal{A}) = \bigoplus_{n\le r}(s^{-1}\bar{\mathcal{A}})^{\otimes n}, \qquad F_{>r}\bar{B}(\mathcal{A}) = \bigoplus_{n>r}(s^{-1}\bar{\mathcal{A}})^{\otimes n}.$$

The truncation $\Theta^{\le r}_{\mathcal{A}}$ is the projection of $\Theta_{\mathcal{A}}$ modulo $F_{>r}$. The Maurer–Cartan recursion determines each new degree up to a coboundary from the previous truncation.

Definition 2.1 (Shadow obstruction tower)**.** The *degree-r shadow* of $\mathcal{A}$ is

$$\mathrm{Sh}_r(\mathcal{A}) \;:=\; -\nabla_H^{-1}\big([\Theta^{\le r-1}_{\mathcal{A}}, \Theta^{\le r-1}_{\mathcal{A}}]^{(0)}_r\big) \;\in\; \mathcal{A}^{\mathrm{sh}}_{r,0},$$

where the bracket is the genus-zero graph composition, the subscript r extracts the degree-r component, and ∇_H is the Hessian covariant derivative on the shadow algebra $\mathcal{A}^{\mathrm{sh}} = H^*(\mathfrak{g}^{\mathrm{mod},(0)}_{\mathcal{A}})$ descended from the convolution Lie bracket. The sequence $(\mathrm{Sh}_r(\mathcal{A}))_{r\ge 2}$ is the *shadow obstruction tower* of $\mathcal{A}$.

The definition is bar-intrinsic: if $\Theta_{\mathcal{A}} \equiv \Theta'_{\mathcal{A}}$ mod gauge, then $\mathrm{Sh}_r(\mathcal{A})$ and $\mathrm{Sh}'_r(\mathcal{A})$ coincide as cohomology classes in $\mathcal{A}^{\mathrm{sh}}_{r,0}$.

Remark 2.2 (Closed formulas). On the scalar projection, writing $\Theta_{\mathcal{A}} = \kappa \cdot \eta \otimes \Lambda$ and $H(t) = \sum_{r\ge 2} rS_r t^r$, one has $\mathrm{Sh}_2 = \kappa$, $\mathrm{Sh}_3 = \mathfrak{C}$, $\mathrm{Sh}_4 = \mathfrak{Q}$, and the shadow metric identity

$$H(t)^2 = t^4\big(4\kappa^2 + 12\kappa\mathfrak{C}\, t + (9\mathfrak{C}^2 + 16\kappa\mathfrak{Q})\, t^2\big)$$

on the families where the Riccati reduction has been proved.

3. The L_∞-formality obstruction tower

3.1. **Homotopy transfer on a strict model.** Let $\mathfrak{g}$ be a strict dg Lie algebra with differential d and bracket $[-,-]$. A *deformation retract* onto $H^*(\mathfrak{g})$ is a triple (ι, π, h) with $\iota\colon H^*(\mathfrak{g}) \to \mathfrak{g}$, $\pi\colon \mathfrak{g} \to H^*(\mathfrak{g})$, and $h\colon \mathfrak{g} \to \mathfrak{g}$ of degree -1, satisfying $\pi\iota = \mathrm{id}$ and $\mathrm{id} - \iota\pi = [d, h]$. The homotopy transfer theorem of Kadeishvili [2] and Kontsevich–Soibelman [4], in the form of Loday–Vallette [5, §10.3], produces a transferred L_∞-algebra structure $(H^*(\mathfrak{g}), 0, \{\ell^{\mathrm{tr}}_n\}_{n\ge 2})$ quasi-isomorphic to $\mathfrak{g}$, with brackets computed by the Kadeishvili–Merkulov tree sum

$$\ell^{\mathrm{tr}}_r(a_1, \dots, a_r) \;=\; \sum_{T\in \mathsf{Trees}_r} \frac{\pm 1}{|\mathrm{Aut}(T)|}\; \pi \circ \Phi_T(\iota(a_1), \dots, \iota(a_r)), \tag{3.1}$$

where Trees_r is the set of planar rooted trees with r leaves and Φ_T decorates internal vertices with $[-,-]$ and internal edges with h.

3.2. **The formality obstruction tower.**

Definition 3.1 (L_∞-formality and obstructions)**.** The dg Lie algebra $\mathfrak{g}$ is *L_∞-formal* if there is an L_∞ quasi-isomorphism $\mathfrak{g} \simeq (H^*(\mathfrak{g}), 0, [-,-])$; equivalently, $\ell^{\mathrm{tr}}_n = 0$ in L_∞ cohomology for all $n \ge 3$. The *L_∞-formality obstruction tower* is the sequence

$$\big(o_r(\mathfrak{g})\big)_{r\ge 3}, \qquad o_r(\mathfrak{g}) \;:=\; [\ell^{\mathrm{tr}}_r] \;\in\; H^*\big(\mathrm{CE}^{\bullet}_{\mathrm{cyc}}(\mathfrak{g})\big).$$

If $o_r(\mathfrak{g}) = 0$, an L_∞ homotopy kills ℓ^{tr}_r and the next obstruction is defined.

For a pronilpotent dg Lie algebra with Maurer–Cartan element Θ, evaluating the transferred brackets on Θ produces a refined tower $(\ell_r^{\mathrm{tr}}(\Theta, \dots, \Theta))_{r\geq 2}$, which takes values in the convolution target. Obstruction cohomology records the projected class. The refined target-valued tower is the one compared with the shadow tower.

Remark 3.2 (Cyclic deformation complex). For $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$, the relevant deformation complex is the cyclic Chevalley complex of the genus-zero convolution dg Lie algebra. Under the chirally Koszul hypotheses and the chosen cyclic contraction, this complex maps to the shadow cohomology $\mathcal{A}^{\mathrm{sh}}_{*,0}$ by the projection π.

4. Base cases: degrees 2, 3, 4

Proposition 4.1 (Shadow–formality identification at low degree). *For every verified chirally Koszul standard algebra in the fixed genus-zero cyclic transfer setting, the following identities hold:*

(i) $\mathrm{Sh}_2(\mathcal{A}) \;=\; \kappa(\mathcal{A}) \;=\; \ell_2^{(0)}(\pi\Theta_{\mathcal{A}}, \pi\Theta_{\mathcal{A}})$.

(ii) $\mathrm{Sh}_3(\mathcal{A}) \;=\; \mathfrak{C}(\mathcal{A}) \;=\; -h\big(\ell_3^{(0)}(\pi\Theta_{\mathcal{A}}^{\leq 2}, \pi\Theta_{\mathcal{A}}^{\leq 2}, \pi\Theta_{\mathcal{A}}^{\leq 2})\big)$.

(iii) $\mathrm{Sh}_4(\mathcal{A}) \;=\; \mathfrak{Q}(\mathcal{A}) \;=\; \big[\ell_4^{(0),\mathrm{tr}}(\pi\Theta_{\mathcal{A}}^{\leq 3}, \pi\Theta_{\mathcal{A}}^{\leq 3}, \pi\Theta_{\mathcal{A}}^{\leq 3}, \pi\Theta_{\mathcal{A}}^{\leq 3})\big]$.

Proof. Part (i). The binary bracket on $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$ is the genus-zero convolution Lie bracket; evaluating on $\Theta_{\mathcal{A}}$ produces the scalar curvature $\kappa(\mathcal{A}) \in \mathbb{Q}(c)$ by definition [6, Theorem D]. The identity $\mathrm{Sh}_2(\mathcal{A}) = \kappa(\mathcal{A})$ is the degree-2 scalar shadow, so the three quantities agree.

Part (ii). The ternary bracket $\ell_3^{(0)}$, computed by (3.1), projects after cyclic symmetrisation to the three s-, t-, and u-channel stable four-point trees. On the primary line $[\ell_3^{(0)}] = 0$ since $H^1(\overline{\mathcal{M}}_{0,4}) = H^1(\mathbb{P}^1) = 0$; at the chain level $\ell_3^{(0)}$ is exact and nonzero, and homotopy transfer gives $\mathfrak{C}(\mathcal{A}) = -h(\ell_3^{(0)}(\pi\Theta_{\mathcal{A}}^{\leq 2}, \pi\Theta_{\mathcal{A}}^{\leq 2}, \pi\Theta_{\mathcal{A}}^{\leq 2}))$. The cubic shadow $\mathfrak{C}(\mathcal{A})$, computed as the three-channel graph sum at degree three, matches the cyclic projection of the transferred tree sum with identical weights and propagators.

Part (iii). The quaternary bracket $\ell_4^{(0)}$ is a planar HPL tree sum whose cyclic projection is indexed by the 15 labelled genus-zero stable trees with five external legs. Its transferred class $\ell_4^{(0),\mathrm{tr}}$ lies in $H^*(\overline{\mathcal{M}}_{0,5})$. The weight-four graph sum defining $\mathfrak{Q}(\mathcal{A})$ enumerates those labelled stable trees with matching weights, signs, and propagators [6, appendix on nonlinear modular shadows], so $\mathrm{Sh}_4(\mathcal{A}) = \mathfrak{Q}(\mathcal{A}) = [\ell_4^{(0),\mathrm{tr}}(\pi\Theta_{\mathcal{A}}^{\leq 3}, \dots, \pi\Theta_{\mathcal{A}}^{\leq 3})]$. □

Remark 4.2 (Degree-2: κ is family-dependent). The identity $\mathrm{Sh}_2(\mathcal{A}) = \kappa(\mathcal{A})$ holds at degree two on the scalar projection for the verified standard families; on the primary line this is the definition $S_2 := \kappa$. What is family-dependent is the value of κ itself: $\kappa^{\mathrm{Vir}} = c/2$, $\kappa^{\mathrm{Heis}} = k$, $\kappa^{\mathrm{KM}} = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$, $\kappa^{\mathcal{W}_N} = c(H_N - 1)$ with $H_N = \sum_{j=1}^{N} 1/j$. The formula $\kappa = c/2$ is Virasoro-specific.

5. The comparison proof

Theorem 5.1 (Genus-zero shadow–formality comparison). *Let $\mathcal{A}$ be a verified chirally Koszul standard algebra in the genus-zero cyclic setting, with fixed cyclic contraction (ι, π, h) of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$ onto $\mathcal{A}^{\mathrm{sh}}_{*,0}$. Assume the cyclic stable-tree/HPL comparison lemma. For every degree $r \geq 2$,*

$$\mathrm{Sh}_r(\mathcal{A}) \;=\; \ell_r^{(0),\mathrm{tr}}\big(\pi\Theta_{\mathcal{A}}^{\leq r-1}, \dots, \pi\Theta_{\mathcal{A}}^{\leq r-1}\big) \quad \text{in } \mathcal{A}^{\mathrm{sh}}_{r,0}, \tag{5.1}$$

where $\ell_r^{(0),\mathrm{tr}}$ is the transferred genus-zero L_∞ bracket on the minimal model of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$. For $r = 2$ this is the curvature input. For $r \geq 3$ it is the formality obstruction evaluation on the truncated Maurer–Cartan element.

Proof. Since genus-zero stable graphs are trees, $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod},(0)}$ is a strict dg Lie algebra. The fixed cyclic contraction satisfies $\pi\iota = \mathrm{id}$, $\mathrm{id} - \iota\pi = [D, h]$. The homotopy transfer theorem [2, 4, 5] produces a transferred L_∞ structure $(\mathcal{A}^{\mathrm{sh}}_{*,0}, \{\ell_r^{(0),\mathrm{tr}}\})$, with brackets computed by (3.1).

At $r = 2$, both sides equal $\kappa(\mathcal{A})$ by Proposition 4.1(i); the cases $r = 3, 4$ are parts (ii), (iii). For $r \geq 5$, assume (5.1) through degree $r - 1$. The cyclic stable-tree/HPL comparison lemma identifies the graph sum for $\mathrm{Sh}_r(\mathcal{A})$ with the Kadeishvili–Merkulov tree formula for $\ell_r^{(0),\mathrm{tr}}(\pi\Theta^{\leq r-1}, \ldots, \pi\Theta^{\leq r-1})$: internal edges carry h, outputs are projected by π, and internal vertices are the lower-degree transferred brackets. The projected sums agree in $\mathcal{A}^{\mathrm{sh}}_{*,0}$. □

Remark 5.2 (Genus-zero projection and positive-genus terms)*.* Theorem 5.1 concerns the genus-zero projection. For scalar uniform-weight families the full shadow tower has the expansion

$$\mathrm{Sh}_r(\mathcal{A}) \;=\; \underbrace{\ell_r^{(0),\mathrm{tr}}(\pi\Theta_{\mathcal{A}}^{\leq r-1}, \ldots)}_{\text{genus-zero formality}} \;+\; \underbrace{\sum_{g\geq 1} \ell_r^{(g),\mathrm{tr}}(\ldots) + \Lambda_P(\ldots)}_{\text{genus}\ \geq 1\ \text{corrections}},$$

where Λ_P is the BV loop operator. The first summand is controlled by classical homotopy transfer. The positive-genus summands are governed by the quantum L_∞ operations and the BV loop operator. At genus $g \geq 2$ with multi-weight generators, the scalar formula for the genus corrections receives a cross-channel term $\delta F_g^{\mathrm{cross}}$; the genus-zero comparison (5.1) is unchanged.

6. Common Depth

Definition 6.1 (Two genus-zero depth invariants)**.** For a chiral algebra $\mathcal{A}$, set $r_{\max}(\mathcal{A}) := \sup\{r \geq 2 : \mathrm{Sh}_r(\mathcal{A}) \neq 0\}$ (shadow termination degree), and $f_\infty^{\Theta}(\mathcal{A}) := \sup\{n \geq 2 : \ell_n^{(0),\mathrm{tr}}(\pi\Theta_{\mathcal{A}}^{\leq n-1}, \ldots, \pi\Theta_{\mathcal{A}}^{\leq n-1}) \neq 0\}$ (evaluated L_∞ obstruction level on the transferred genus-zero model).

Corollary 6.2 (Evaluated common depth for the standard cyclic families)**.** *For the verified chirally Koszul standard families in the genus-zero cyclic setting,*

$$r_{\max}(\mathcal{A}) = f_\infty^{\Theta}(\mathcal{A}).$$

Proof. The equality is the evaluation identity of Theorem 5.1, degree by degree. Operation-level vanishing requires an independent nondegeneracy lemma for the transferred cyclic primitive component. □

Corollary 6.3 (Coarse depth realisation)**.** *Within the constructed standard cyclic families, the four coarse shadow-depth classes are all realized: family:*

$$\boldsymbol{G}\,(r_{\max} = 2), \quad \boldsymbol{L}\,(r_{\max} = 3), \quad \boldsymbol{C}\,(r_{\max} = 4), \quad \boldsymbol{M}\,(r_{\max} = \infty).$$

The class $\boldsymbol{G}$ *consists of* L_∞*-formal algebras; the class* $\boldsymbol{M}$ *consists of intrinsically non-formal algebras, in the sense that no finite truncation of their minimal* L_∞ *model is quasi-isomorphic to a dg Lie algebra.*

Proof. Theorem 5.1 identifies shadow depth with the L_∞ obstruction level after evaluation on the transferred Maurer–Cartan element. The examples in Section 7 realize each coarse class. □

7. Examples

The four standard families realize the four shadow-depth classes.

Example 7.1 (Heisenberg; class **G**, formal)**.** The Heisenberg vertex algebra Heis_k at level k, with $J(z)J(w) \sim k/(z-w)^2$, has $\kappa(\mathrm{Heis}_k) = k$. The bar differential is strictly commutative, so $m_n = 0$ and $\ell_n^{(0),\mathrm{tr}} = 0$ for $n \geq 3$. Hence $r_{\max}(\mathrm{Heis}_k) = 2$ and Heis_k is L_∞-formal. Class **G**.

Example 7.2 (Affine Kac–Moody; class **L**). At non-critical level $k \neq -h^\vee$, the affine Kac–Moody vertex algebra $\widehat{\mathfrak{g}}_k$ has $\kappa(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$. The binary bracket is the Lie bracket on $\mathfrak{g}$; Jacobi kills all degree-four graph contributions, so $\ell_4^{(0),\mathrm{tr}} = 0$, while $\ell_3^{(0),\mathrm{tr}} \neq 0$ in general. Hence $r_{\max}(\widehat{\mathfrak{g}}_k) = 3$. Class **L**.

Example 7.3 ($\beta\gamma_\lambda$; class **C**, shadow depth 4). The standard rank-one $\beta\gamma_\lambda$ contact family has

$$S_2 = 6\lambda^2 - 6\lambda + 1, \qquad S_3 = 0, \qquad S_4 = -\frac{5}{12}, \qquad S_r = 0 \quad (r \geq 5).$$

Thus $r_{\max}(\beta\gamma_\lambda) = 4$ at the generic contact points. Equivalently, the bracket $\ell_3^{(0),\mathrm{tr}}$ vanishes and $\ell_4^{(0),\mathrm{tr}}$ is nonzero. Class **C**.

Example 7.4 (Virasoro; class **M**, obstructed at every degree). The Virasoro vertex algebra Vir_c has $\kappa(\mathrm{Vir}_c) = c/2$. The transferred brackets on the primary line of $H^*(\bar{B}(\mathrm{Vir}_c))$ satisfy $m_k^{\mathrm{tr}}(s^{-1}T, \dots, s^{-1}T) = S_k \cdot e_{2k}$ for $k \geq 2$, with $s^{-1}T$ the desuspension of the conformal vector to bar degree one and e_{2k} the weight-$2k$ basis vector of Ext^1. The first shadow coefficients are

$$S_2 = \tfrac{c}{2}, \qquad S_3 = 2\ (c\text{-independent}), \qquad S_4 = \tfrac{10}{c(5c+22)},$$
$$S_5 = -\tfrac{48}{c^2(5c+22)}, \qquad S_6 = \tfrac{4(240c+1031)}{c^3(5c+22)^2}.$$

The weighted Riccati-metric coefficients $\widehat{S}_6^{wt}$ and $\widehat{S}_7^{wt}$ use a different normalization. Here Ext^1 denotes the primary bar-cohomology line in the transferred shadow model. It is not a Hochschild-degree statement; Theorem H is the separate chiral Hochschild concentration theorem in degrees $\{0, 1, 2\}$, with symmetric descent obtained by averaging. For generic c on the nonsingular Virasoro shadow surface, the Riccati recurrence and computed coefficients give infinitely many nonzero S_k. Thus $r_{\max}(\mathrm{Vir}_c) = \infty$. The normal-ordered composites are the source of possible operations, not the nonvanishing proof. Class **M**.

Remark 7.5 (The four standard depths). For the verified finite-type standard families governed by the quadratic Riccati reduction, the coarse shadow-depth classes are **G**, **L**, **C**, and **M**. The identity $H(t)^2 = t^4Q(t)$ with $\deg Q \leq 2$ gives depths 2, 3, 4, and ∞ for this scalar projection [6, theorem on Riccati algebraicity]. This is a coarse classification of the standard cyclic families. Fine finite-depth towers require separate analysis.

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